Title

Your name here

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Exercise 1 [2]

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Tools and resources used: On Exercise 1, I used ChatGPT to help me figure out how to [...]. On Choice Exercise 7, I watched a YouTube video to [...].

Exercise 1 Points attempted: 2. Time spent: 5 minutes. Difficulty: 1/10. Self-assessed score: 2. Questions/Feedback: None.

Exercise 2 Points attempted: 1. Time spent: 2 minutes. Difficulty: 1/10. Self-assessed score: 1. Questions/Feedback: None.

Exercise 3 Points attempted: 6. Time spent: 40 minutes. Difficulty: 3/10. Self-assessed score: 6. Questions/Feedback: Did I do the problem correctly and as intended?

Exercise 4 Points attempted: 5. Time spent: 35 minutes. Difficulty: 4/10. Self-assessed score: 5. Questions/Feedback: None.

Exercise 5 Points attempted: 1. Time spent: 5 minutes. Difficulty: 1/10. Self-assessed score: 1. Questions/Feedback: None.

Exercise 8 Points attempted: 7. Time spent: 40 minutes. Difficulty: 4/10. Self-assessed score: 7. Questions/Feedback: Mathematical correctness.

Exercise 2 [1]

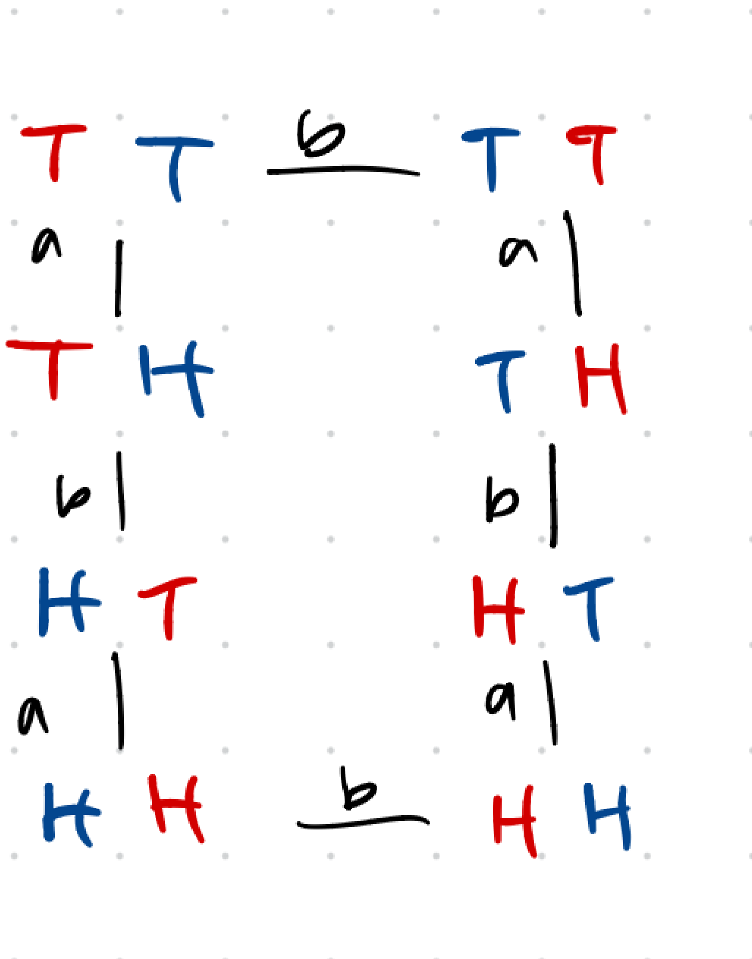
Done! Posted an image of a marker with rotational symmetry.

Exercise 3 [6]

1. The coins have 8 symmetries. That is because there are 2^2 configurations without swapping places. When allowing swapping places, we get $2 \times 2^2 = 8$.

2. First, I chose the symmetry of flipping the *left*, denoted as f_L, coin over. Applying this symmetry twice gets us back to id. However, when we allow the swapping of places, denoted s, we can actually get to all states. We do not actually need an explicit 'flip right' symmetry, since it is actually equal to $f_L \circ s$.

3. In the diagram, let $a = f_L$ and $b = s$.



4. I used 2 generators.

5. This group is not commutative. This is because $f_L \circ s \neq s \circ f_L$. Swapping the coins then flipping the left is not the same as flipping then swapping.

Exercise 4 [5]

1. First, we write down the numbers in order to see 123456. Now, lets say we want to swap some things around, so we move some things around. Imagine that we move 1 to 2 and 2 to 1. We would get 213456. If we want to make some more complicated moves, we can actually send any number anywhere! So, $\pi = 152634$ would send 1 to 1 (unchanged), 2 to 5, 3 to 2, 4 to 6, 5 to 3, and 6 to 4.

2. We would get $\pi \circ \pi = 135426$.

3. We would get $(\pi \circ \pi) \circ \pi = 123654$.

4. Since $\pi^3 = 123654$, which is really just swapping 4 and 6, $\pi^3 \circ pi^3 = \pi^6 = 123456$. So, π has an order of 6.

5. The right inverse of π is $\sigma = 135624$.

6. The left inverse of π is $\sigma = 135624$.

7. A word with order 1 is just any of the identity permutations. For example, 1234. For a word with order 2, we have 21. A general formula for constructing n order words is to just have n items, and for the word to be $\pi = 2345 ... n1$. This will "cycle" the items. So for order 4, we have 2341.

Exercise 5 [1]

1. The symmetry $r * h$ is a 120 degree rotation followed by a horizontal reflection.
2. The symmetry $r \star h$ is a 120 degree rotation after a horizontal reflection.
3. The function $f \circ g$ behaves like f after h.

Exercise 8 [7]

1. [1] Write the word $2716543 \in S_7$ in cycle notation.

   The word $2716543 \in S_7$ is equal to (2731)(64) in cycle notation.

2. [1] Write the permutation $(3)(514)(726)(8) \in S_8$ as a word.

   The permutation $(514)(726) \in S_8$ is equal to the word 46351728.

3. [1] Let $\pi = (514)(726)$ and let $\sigma = (618)(45)$. Write $\pi * \sigma$ in cycle notation, where $*$ means "followed by"?

   Let $\pi = (514)(726)$ and let $\sigma = (618)(45)$.

   $$\pi * \sigma = (586721).$$

4. [1] Recall that the order of an element is the smallest positive number of times we have to apply it to get back to the identity. The order of the element $\tau = (123)(456)(7) \in S_7$ is 3 because $\tau \neq e \neq \tau^2$, but $\tau^3 = e$ What is the order of the element $(51)(632) \in S_7$?

   The order of the element $\tau = (123)(456)(7) \in S_7$ is 3 because $\tau \neq e \neq \tau^2$, but $\tau^3 = e$ What is the order of the element $(51)(632) \in S_7$?

5. [1] Can you find an element in S_5 that has order > 5? If so, give an example (written in cycle notation). If not, explain why not.

   Yes. The element (12)(345) has an order of 6.

6. [2] What is the greatest order an element in S_7 can have? In S_{10}?

   The greatest order of an element in S_7 is (maybe) 12. This is done by making an element with two cycles, one with "size" 3 and order 3 and another cycle with "size" 4 and order 4. Then LCM(3, 4) = 12. I am actually unsure of whether this is the true max value for the order of elements in S_7, since that would seemingly require checking many subcases.

   For elements of S_{10}, I believe the max value is 30. In doing this problem, I decided to write a Python script to brute force all combos. The result ended up being 30.

```python
from math import lcm


def get_distinct_sums(target, start=1):
    # Base case: if target is 0, we found a valid combination
    if target == 0:
        return [[]]

    results = []

    # Iterate from 'start' to the target
    for i in range(start, target + 1):
        # Recursively find combinations for the remaining value
        # We pass 'i + 1' to ensure the next number is strictly greater (distinct)
        sub_combinations = get_distinct_sums(target - i, i + 1)

        for combo in sub_combinations:
            results.append([i] + combo)

    return results

s7_sizes = get_distinct_sums(7)
```

```python
s7_max_order = max(lcm(*size) for size in s7_sizes)
print(s7_max_order)


s10_sizes = get_distinct_sums(10)
s10_max_order = max(lcm(*size) for size in s10_sizes)
print(s10_max_order)
```

Choice Exercise 10 [3 + 3]

In class, we found that we can represent 120° rotation of the plane (one of the dihedral symmetries) with the matrix

$$R = \begin{pmatrix} -\frac{1}{2} & -\frac{\sqrt{3}}{2} \\ \frac{\sqrt{3}}{2} & -\frac{1}{2} \end{pmatrix}.$$

1. [3] Write down the matrix that corresponds to f, a left-to-right flip, and then write down the matrices for the remaining four dihedral symmetries of the triangle D_3.

   We write the matrix F to represent the 'left-to-right flip' as:

   $$F = \begin{pmatrix} -1 & 0 \\ 0 & 1 \end{pmatrix}.$$

   First, we also define the id symmetry as the identity matrix:

   $$I = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}.$$

   Then, we can construct all other symmetries of D_3 by composing F with R, meaning we take the matrix products of F and R. By convention, I will apply the F last.

   So, we also have the symmetries R^2, $F \times R$, and $F \times R^2$.